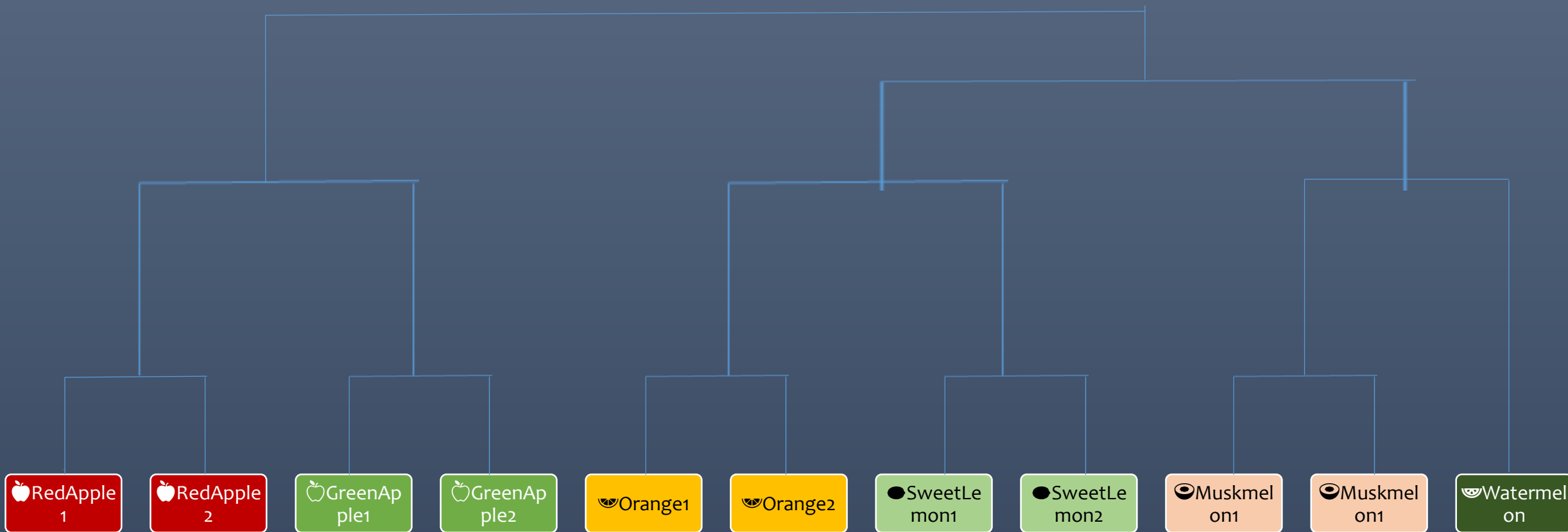
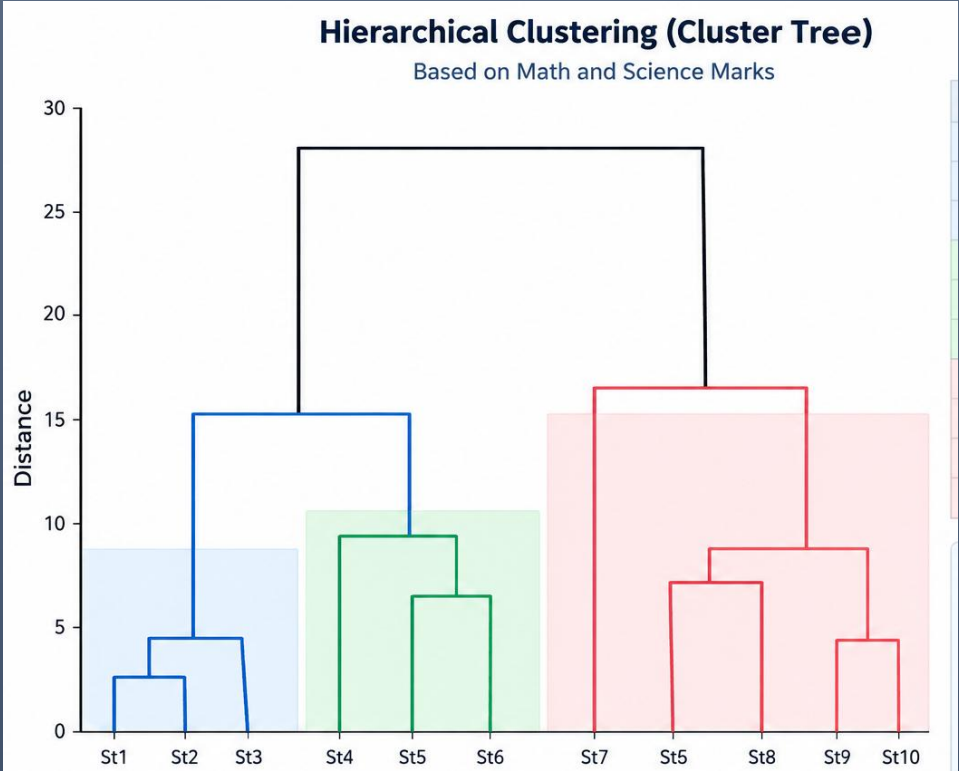


Fruit		Weight (g)	Sweetness
	RedApple1	150	7
	RedApple2	160	8
	GreenApple	155	6
	Orange1	120	6
	Orange2	130	7
	SweetLemon	125	5
	Muskmelon	500	8
	Watermelon	800	7



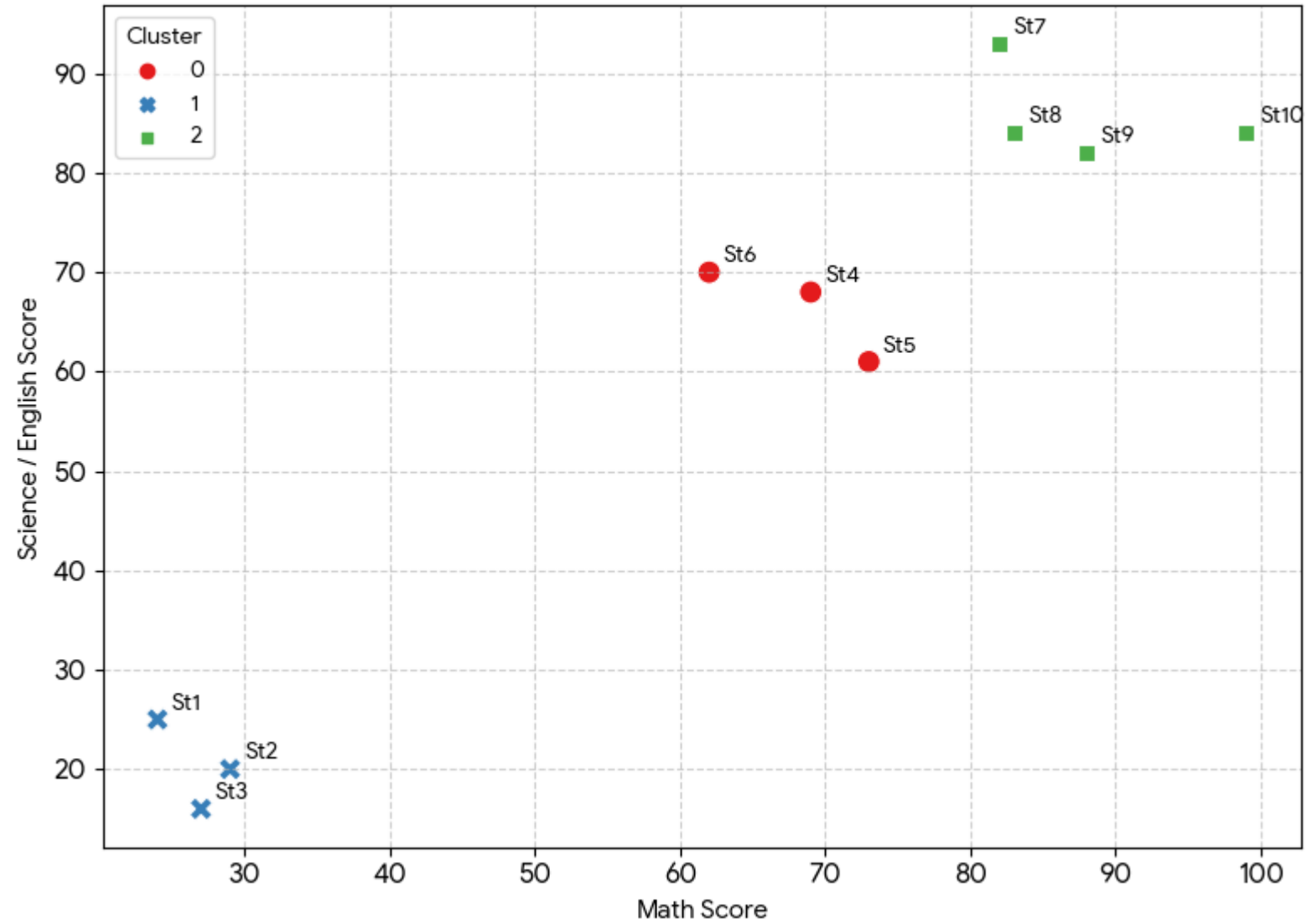
Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84



Cluster 1 (Low Marks) St1, St2, St3	Cluster 2 (Medium Marks) St4, St5, St6	Cluster 3 (High Marks) St7, St8, St9, St10
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Student	Math	Science	Cluster
St1	24	25	1
St2	29	20	1
St3	27	16	1
St4	69	68	2
St5	73	61	2
St6	62	70	2
St7	82	93	3
St8	83	84	3
St9	88	82	3
St10	99	84	3

Cluster Plot: Math vs Science (or English scores from image)



The Euclidean distance between two points (x_1, y_1) and (x_2, y_2) is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

1. St-1 and St-2

$$\begin{aligned}d &= \sqrt{(24 - 29)^2 + (25 - 20)^2} \\&= \sqrt{(-5)^2 + (5)^2} \\&= \sqrt{25 + 25} \\&= \sqrt{50} = 7.07\end{aligned}$$

2. St-1 and St-3

$$\begin{aligned}d &= \sqrt{(24 - 27)^2 + (25 - 16)^2} \\&= \sqrt{(-3)^2 + 9^2} \\&= \sqrt{9 + 81} \\&= \sqrt{90} = 9.49\end{aligned}$$

3. St-1 and St-4

$$\begin{aligned}d &= \sqrt{(24 - 69)^2 + (25 - 68)^2} \\&= \sqrt{(-45)^2 + (-43)^2} \\&= \sqrt{2025 + 1849} \\&= \sqrt{3874} = 62.24\end{aligned}$$

4. St-1 and St-5

$$\begin{aligned}d &= \sqrt{(24 - 73)^2 + (25 - 61)^2} \\&= \sqrt{(-49)^2 + (-36)^2} \\&= \sqrt{2401 + 1296} \\&= \sqrt{3697} = 60.80\end{aligned}$$

5. St-1 and St-6

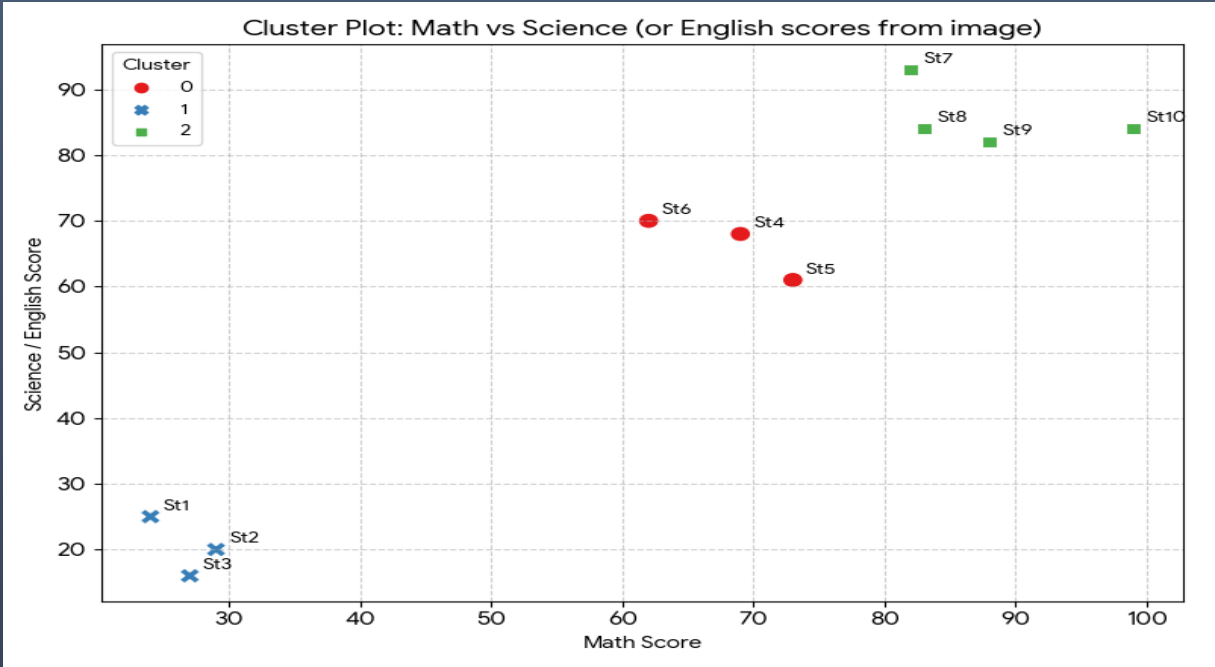
$$\begin{aligned}d &= \sqrt{(24 - 62)^2 + (25 - 70)^2} \\&= \sqrt{(-38)^2 + (-45)^2} \\&= \sqrt{1444 + 2025} \\&= \sqrt{3469} = 58.89\end{aligned}$$

Student	To $C_1(\text{St-1})$
St-1	-
St-2	7.1
St-3	9.5
St-4	62.2
St-5	60.8
St-6	58.9
St-7	89.4
St-8	83.4
St-9	85.7
St-10	95.4

Student	To $C_2(\text{St-10})$
St-1	95.4
St-2	94.9
St-3	99.0
St-4	34.0
St-5	34.7
St-6	39.6
St-7	19.2
St-8	16.0
St-9	11.2
St-10	-

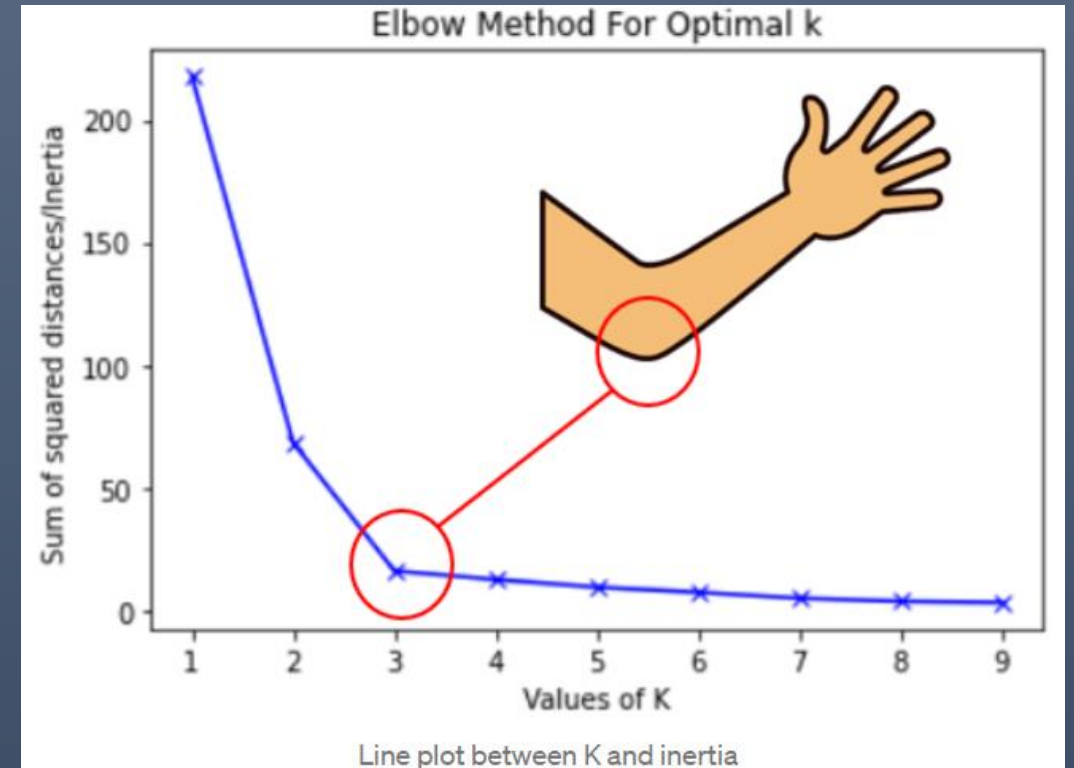
Smaller Distance
-
7.1
9.5
34.0
34.7
39.6
19.2
16.0
11.2
-

Distance ²
-
50.0
90.0
1,156.0
1,205.0
1,565.0
370.0
256.0
125.0
-



EL-Bow:

WCSS-Within cluster sum of squared distance



Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

Student	St-1	St-2	St-3		St-4	St-5	St-6		St-7	St-8	St-9	St-10
St-1	-	7.1	9.2		62.2	64.9	58.0		89.5	83.5	85.4	94.6
St-2	7.1	-	5.0		62.2	63.1	59.0		89.8	84.0	85.4	91.6
St-3	9.2	5.0	-		66.5	67.3	63.5		93.3	88.3	89.2	96.1
St-4	62.2	62.2	66.5		-	7.8	7.3		27.8	20.5	24.4	32.0
St-5	64.9	63.1	67.3		7.8	-	14.9		33.3	23.2	20.6	26.0
St-6	58.0	59.0	63.5		7.3	14.9	-		31.9	23.7	30.0	38.0
St-7	89.5	89.8	93.3		27.8	33.3	31.9		-	9.1	10.3	18.4
St-8	83.5	84.0	88.3		20.5	23.2	23.7		9.1	-	5.8	16.3
St-9	85.4	85.4	89.2		24.4	20.6	30.0		10.3	5.8	-	11.4
St-10	94.6	91.6	96.1		32.0	26.0	38.0		18.4	16.3	11.4	-

Group1:St-1,St-2,St-3

Group2:St-4,St-5,St-6

Group3:St-7,St-8,St-9,St-10

Student	Math	Science	Cluster
St1	24	25	1
St2	29	20	1
St3	27	16	1
St4	69	68	2
St5	73	61	2
St6	62	70	2
St7	82	93	3
St8	83	84	3
St9	88	82	3
St10	99	84	3